Minimal Coupling and the Maxwell–Dirac Equations

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1 A quick review on principal bundles

This section fixes conventions and recalls the minimal principal-bundle technology needed later for gauge theory and minimal coupling.

1.1 Principal bundles and fundamental vector fields

Let G be a Lie group with Lie algebra $\mathfrak{g} = \mathrm{Lie}(G)$. A (right) *principal G-bundle* over a smooth manifold M is a smooth surjective submersion $\pi : P \to M$ together with a smooth right action $R_g : P \to P$, $p \mapsto p \cdot g$, such that

- the action is free and proper, and
- $\pi(p \cdot g) = \pi(p)$ and the action is transitive on each fiber $P_x = \pi^{-1}(x)$.

Equivalently, each fiber P_x is a G-torsor.

Remark 1.1 (Right-action convention)**.** Throughout, the structure group acts on the *right*. This determines the signs in gauge transformation laws below. If you prefer left actions, the formulas must be adjusted systematically.

Denote the *vertical subbundle* by

$$V := \ker(\mathrm{d}\pi) \subset TP.$$

For $\xi \in \mathfrak{g}$, the associated *fundamental vector field* ξ_P on P is

$$\xi_P(p) \;:=\; \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} p \cdot \exp(t\xi) \in T_pP.$$

This identifies each vertical space with the Lie algebra:

$$V_p \cong \mathfrak{g}, \qquad \xi \longleftrightarrow \xi_P(p).$$

1.2 Principal connections: distributions and connection forms

There are two equivalent viewpoints that are both useful later.

Definition 1.2 (Ehresmann principal connection)**.** A *principal connection* on $\pi : P \to M$ is a smooth subbundle $H \subset TP$ (the *horizontal distribution*) such that:

1. $TP = H \oplus V$ (direct sum decomposition), and
2. H is G-equivariant: $(\mathrm{d}R_g)_p(H_p) = H_{p\cdot g}$ for all $g \in G$.

Given $x \in M$, $p \in P_x$, and $X \in T_xM$, there is a unique *horizontal lift* $X_p^{\mathrm{hor}} \in H_p$ satisfying $\mathrm{d}\pi(X_p^{\mathrm{hor}}) = X$.

Definition 1.3 (Connection 1-form)**.** A *(principal) connection 1-form* is a $\mathfrak{g}$-valued 1-form $\omega \in \Omega^1(P;\mathfrak{g})$ such that:

1. (**reproduction**) $\omega(\xi_P) = \xi$ for all $\xi \in \mathfrak{g}$,
2. (G-**equivariance**) $(R_g)^*\omega = \mathrm{Ad}(g^{-1})\,\omega$ for all $g \in G$.

Proposition 1.4 (Equivalence of the two definitions)**.** *Given a principal connection form ω, the horizontal distribution $H := \ker\omega$ is a principal connection in the Ehresmann sense. Conversely, given a principal horizontal distribution H, there is a unique $\mathfrak{g}$-valued 1-form ω satisfying the two axioms above and $\ker\omega = H$.*

Remark 1.5 (How to think about ω geometrically)**.** At each $p \in P$, the form $\omega_p : T_pP \to \mathfrak{g}$ is the projection onto the vertical part $V_p \cong \mathfrak{g}$ along H_p. In particular, a tangent vector is horizontal iff ω annihilates it.

1.3 Curvature of a principal connection

To write the curvature formula, recall the standard bracket-wedge operation for $\mathfrak{g}$-valued forms. For $\alpha \in \Omega^p(P;\mathfrak{g})$ and $\beta \in \Omega^q(P;\mathfrak{g})$, one defines a $(p+q)$-form $[\alpha \wedge \beta] \in \Omega^{p+q}(P;\mathfrak{g})$ by combining the wedge product on forms with the Lie bracket $[\cdot,\cdot]$ on $\mathfrak{g}$. In particular, for $p = q = 1$,

$$[\alpha \wedge \beta](X,Y) = [\alpha(X), \beta(Y)] - [\alpha(Y), \beta(X)].$$

Definition 1.6 (Curvature 2-form)**.** Given a connection form $\omega \in \Omega^1(P;\mathfrak{g})$, its *curvature* is the $\mathfrak{g}$-valued 2-form

$$\Omega \;:=\; \mathrm{d}\omega \;+\; \tfrac{1}{2}[\omega \wedge \omega] \;\in\; \Omega^2(P;\mathfrak{g})\,.$$

Equivalently, for $X, Y \in T_pP$,

$$\Omega(X,Y) = \mathrm{d}\omega(X,Y) + [\omega(X), \omega(Y)].$$

The curvature has two key structural properties:

- **horizontality:** Ω vanishes if one input is vertical,
- G**-equivariance:** $(R_g)^*\Omega = \mathrm{Ad}(g^{-1})\,\Omega$.

Therefore, Ω descends to a 2-form on the base with values in the *adjoint bundle*

$$\mathrm{ad}(P) := P \times_{\mathrm{Ad}} \mathfrak{g} \longrightarrow M.$$

Concretely, one obtains $F \in \Omega^2(M;\mathrm{ad}(P))$ whose pullback to P along π recovers Ω.

Remark 1.7 (Bianchi identity)**.** Define the covariant exterior derivative on $\mathfrak{g}$-valued forms by $d_\omega\alpha := \mathrm{d}\alpha + [\omega \wedge \alpha]$ (on suitable horizontal, equivariant forms). Then the curvature satisfies

$$d_\omega \Omega = 0.$$

1.4 Associated bundles

Let $\rho : G \to GL(V)$ be a (finite-dimensional) complex or real representation. The *associated vector bundle* is

$$E := P \times_\rho V \;:=\; (P \times V)/\sim, \qquad (p,v) \sim (p \cdot g, \rho(g^{-1})v).$$

We write $[p,v] \in E$ for the equivalence class of (p,v).
A very efficient way to work with sections is via equivariant maps.

Lemma 1.8 (Sections as equivariant maps)**.** *A section $\sigma \in \Gamma(E)$ is equivalent to a smooth map $f : P \to V$ satisfying the ρ-equivariance condition*

$$f(p \cdot g) = \rho(g^{-1})f(p), \qquad \forall p \in P,\ \forall g \in G,$$

via $\sigma(\pi(p)) = [p, f(p)]$.

1.5 Induced connections on associated bundles

Let ω be a principal connection on P, with horizontal distribution $H = \ker\omega$. We define the induced covariant derivative on any associated bundle $E = P \times_\rho V$ in a coordinate-free way.

Definition 1.9 (Induced covariant derivative via horizontal lifts)**.** Let $\sigma \in \Gamma(E)$ correspond to a ρ-equivariant map $f : P \to V$. For $x \in M$, $X \in T_xM$, and any $p \in P_x$, let $X_p^{\mathrm{hor}} \in H_p$ be the horizontal lift. Define $\nabla^\omega_X\sigma \in E_x$ by requiring it to be represented by the derivative of f along the horizontal lift:

$$(\nabla^\omega_X\sigma)(x) \;\text{ is represented by }\; [p,\ (\mathrm{d}f)_p(X_p^{\mathrm{hor}})].$$

Proposition 1.10 (Well-definedness and functoriality)**.** *The definition above is independent of the choice of $p \in P_x$ and yields a connection ∇^ω on E. Moreover, this construction is functorial in ρ: a G-equivariant linear map of representations induces a bundle map intertwining the corresponding covariant derivatives.*

1.6 Local expressions: gauge potential and transformation law

Let $U \subset M$ be open and let $s : U \to P$ be a local section (a local gauge choice). Define the *local connection* 1*-form* (often called the gauge potential)

$$A := s^*\omega \in \Omega^1(U; \mathfrak{g}) .$$

The corresponding local curvature (field strength) is

$$F := s^*\Omega \;=\; \mathrm{d}A + \tfrac{1}{2}[A \owedge A] \in \Omega^2(U; \mathfrak{g}) .$$

If we change the local section by a map $g : U \to G$ via $s'(x) = s(x) \cdot g(x)$, then

$$A' = \mathrm{Ad}(g^{-1})A + g^{-1}\mathrm{d}g, \qquad F' = \mathrm{Ad}(g^{-1})F. \tag{1}$$

Now let $E = P \times_\rho V$ and write a local section $\sigma \in \Gamma(E|_U)$ as

$$\sigma(x) = [\, s(x),\, v(x)\,], \qquad v : U \to V.$$

Then the induced covariant derivative is given locally by the standard formula

$$\nabla^\omega v \;=\; \mathrm{d}v + \rho_*(A)\, v, \tag{2}$$

where $\rho_* : \mathfrak{g} \to \mathfrak{gl}(V)$ is the induced Lie algebra representation. Under the gauge change $s' = s \cdot g$, one has $v' = \rho(g^{-1})v$, and (1) guarantees that (2) transforms covariantly.

Remark 1.11 (Curvature on associated bundles)**.** The curvature of ∇^ω on E is locally $\rho_*(F)$; globally it is a 2-form with values in $\mathrm{End}(E)$.

Cheat sheet (formulas reused later)

(Connection form)	$\omega(\xi_P) = \xi, \quad (R_g)^*\omega = \mathrm{Ad}(g^{-1})\omega.$
(Curvature)	$\Omega = \mathrm{d}\omega + \tfrac{1}{2}[\omega \owedge \omega].$
(Local gauge potential)	$A = s^*\omega, \quad F = s^*\Omega = \mathrm{d}A + \tfrac{1}{2}[A \owedge A].$
(Gauge transform)	$A' = \mathrm{Ad}(g^{-1})A + g^{-1}\mathrm{d}g, \quad F' = \mathrm{Ad}(g^{-1})F.$
(Associated bundle)	$E = P \times_\rho V, \quad \nabla = \mathrm{d} + \rho_*(A).$

2 Electromagnetism as a $\mathrm{U}(1)$ gauge field

In this section we specialize the principal-bundle formalism of Section 1 to $G = \mathrm{U}(1)$ and interpret a principal $\mathrm{U}(1)$-connection as the electromagnetic gauge field. The key point is:

Gauge potential $\Longleftrightarrow$ *local connection* 1*-form*, *field strength* $\Longleftrightarrow$ *curvature.*

2.1 Lie algebra conventions for $\mathrm{U}(1)$

We use the standard identification

$$\mathrm{U}(1) = \{e^{i\theta} : \theta \in \mathbb{R}\} \subset \mathbb{C}, \qquad \mathfrak{u}(1) = \mathrm{Lie}(\mathrm{U}(1)) = i\mathbb{R}.$$

Since $\mathfrak{u}(1)$ is one-dimensional, it is common (and often convenient) to identify $\mathfrak{u}(1) \simeq \mathbb{R}$ via

$$\iota : \mathbb{R} \xrightarrow{\cong} \mathfrak{u}(1), \qquad a \longmapsto ia,$$

and to silently replace $\mathfrak{u}(1)$-valued forms by real-valued forms using ι^{-1}. We will be explicit about the factor of i whenever it matters (especially when coupling to complex matter fields later).

2.2 Electromagnetic potential as a principal connection

Let $\pi : P \to M$ be a principal $\mathrm{U}(1)$-bundle. A principal connection is a $\mathfrak{u}(1)$-valued 1-form $\omega \in \Omega^1(P; \mathfrak{u}(1))$ satisfying the axioms of Section 1. Choosing a local section (a local gauge) $s : U \to P$ produces the *local gauge potential*

$$A \;:=\; s^*\omega \in \Omega^1(U; \mathfrak{u}(1))\,.$$

Gauge transformations. On U, any other local section has the form $s' = s \cdot g$ for a smooth map $g : U \to \mathrm{U}(1)$. Since $\mathrm{U}(1)$ is abelian, Ad is trivial and the general transformation law (1) simplifies to

$$A' \;=\; A + g^{-1}\mathrm{d}g. \tag{3}$$

Writing $g = e^{i\chi}$ for a real function $\chi : U \to \mathbb{R}$, we have $g^{-1}\mathrm{d}g = i\,\mathrm{d}\chi$, so (3) becomes

$$A' = A + i\,\mathrm{d}\chi \qquad \text{in } \mathfrak{u}(1) = i\mathbb{R}.$$

If we identify $\mathfrak{u}(1) \simeq \mathbb{R}$ by dividing by i, then the same formula reads

$$\underline{\text{(physics-style real potential)}} \qquad \mathcal{A}' = \mathcal{A} + \mathrm{d}\chi,$$

where $A = i\mathcal{A}$.

Remark 2.1 (Local vs global potentials)**.** In general, A is only locally defined because it depends on the choice of local section s. What is globally defined is the principal connection ω on P. The curvature (field strength) below is global and gauge-invariant.

2.3 Field strength as curvature

For $G = \mathrm{U}(1)$ the Lie bracket on $\mathfrak{u}(1)$ is zero. Hence the curvature of ω, defined in Section 1, simplifies to

$$\Omega \;=\; \mathrm{d}\omega \quad \in \Omega^2(P; \mathfrak{u}(1))\,.$$

Pulling back by a local section $s : U \to P$ gives the familiar local formula

$$F \;:=\; s^*\Omega \;=\; \mathrm{d}(s^*\omega) \;=\; \mathrm{d}A \quad \in \Omega^2(U; \mathfrak{u}(1))\,. \tag{4}$$

Because $\mathrm{d}(g^{-1}\mathrm{d}g) = 0$, the curvature F is *gauge-invariant*: $F' = F$ under (3). In the real identification $A = i\mathcal{A}$, $F = i\mathcal{F}$, equation (4) reads

$$\underline{\text{(physics-style real field strength)}} \qquad \mathcal{F} = \mathrm{d}\mathcal{A} \in \Omega^2(U; \mathbb{R}).$$

Homogeneous Maxwell equation as Bianchi identity. Since $\Omega = \mathrm{d}\omega$, we have $\mathrm{d}\Omega = 0$ on P, hence $\mathrm{d}F = 0$ on U. Globally, this is the statement that the electromagnetic field strength is closed:

$$\mathrm{d}F = 0.$$

In standard coordinates later, this encodes the homogeneous Maxwell equations.

Remark 2.2 (Topology: first Chern class)**.** The de Rham cohomology class of the curvature determines the topological type of P. More precisely, after dividing by $2\pi i$, one has

$$\left[\frac{F}{2\pi i}\right] \in H^2_{\mathrm{dR}}(M) \quad \text{is the image of } \; c_1(P) \in H^2(M;\mathbb{Z}) \text{ under } H^2(M;\mathbb{Z}) \to H^2_{\mathrm{dR}}(M).$$

Thus, in general, F need not be exact globally even though $\mathrm{d}F = 0$.

2.4 The trivial bundle and a concrete formula on $M \times \mathrm{U}(1)$

When P is trivial, $P = M \times \mathrm{U}(1)$ with right action $(x,u)\cdot g = (x,ug)$, there is a canonical global section $s_0(x) = (x,1)$. Let $\vartheta \in \Omega^1(\mathrm{U}(1);\mathfrak{u}(1))$ be the (right) Maurer–Cartan form $\vartheta = u^{-1}\mathrm{d}u$. Then any principal U(1)-connection can be written uniquely as

$$\omega \;=\; \mathrm{pr}^*_{\mathrm{U}(1)}\vartheta \;+\; \pi^* A_0, \tag{5}$$

where $A_0 := s_0^*\omega \in \Omega^1(M;\mathfrak{u}(1))$ is the global gauge potential in the global gauge s_0.
A different global gauge $s(x) = (x, g(x))$ corresponds to a map $g : M \to \mathrm{U}(1)$ and yields

$$A \;=\; s^*\omega \;=\; A_0 + g^{-1}\mathrm{d}g,$$

recovering the gauge transformation law (3).

Remark 2.3 (Why this matters for Minkowski space)**.** Later we take $M = \mathbb{R}^4$ (Minkowski space). Since $H^2(\mathbb{R}^4;\mathbb{Z}) = 0$, every principal U(1)-bundle over $\mathbb{R}^4$ is trivial, so a global potential exists in some global gauge. Gauge transformations still remain as changes of global section.

2.5 Maxwell action in coordinate-free form

To write Maxwell's equations from a variational principle, assume (M,g) is oriented and equipped with a Lorentzian metric (later $g = \eta$ on $\mathbb{R}^4$). Using the Hodge star $\star$ and the natural inner product on $\mathfrak{u}(1) \simeq i\mathbb{R}$, the Maxwell Lagrangian density is

$$\mathcal{L}_{\mathrm{Max}}(\omega) \;:=\; -\frac{1}{2}\,\langle F, F\rangle \,\mathrm{vol}_g \;=\; -\frac{1}{2}\,F \wedge \star F,$$

where (in the last equality) we use that $\mathfrak{u}(1)$ is one-dimensional so F behaves like a scalar 2-form up to the fixed identification.
In vacuum, the Euler–Lagrange equation for the potential gives the inhomogeneous Maxwell equation

$$\mathrm{d}\star F = 0,$$

and, together with $\mathrm{d}F = 0$, this yields the full Maxwell system in differential-form language. When coupled to matter (Section 5), the right-hand side becomes a current 3-form J:

$$\mathrm{d}\star F = J.$$

2.6 Charged line bundles and the charge-q character

For later coupling to charged matter, fix an integer $q \in \mathbb{Z}$ and consider the character (group homomorphism)

$$\rho_q : \mathrm{U}(1) \to \mathrm{U}(1) \subset \mathbb{C}^\times, \qquad \rho_q(z) = z^q.$$

(The condition $q \in \mathbb{Z}$ is forced if ρ_q is to be a well-defined homomorphism on U(1); physically one often writes the electric charge as $e\,q$ with e a unit.)
The associated complex line bundle is

$$E_q := P \times_{\rho_q} \mathbb{C} \longrightarrow M.$$

By Section 1, the principal connection ω induces a connection $\nabla^{(q)}$ on E_q. Locally, if $\sigma = [s, v]$ with $v : U \to \mathbb{C}$, then

$$\nabla^{(q)} v \;=\; \mathrm{d}v + (\rho_q)_*(A)\, v.$$

In the real-potential convention $A = i\mathcal{A}$, one has $(\rho_q)_*(A) = iq\,\mathcal{A}$, hence

$$\nabla^{(q)} = \mathrm{d} + iq\,\mathcal{A},$$

which is the standard gauge-covariant derivative acting on charge-q complex fields.

Cheat sheet (abelian simplifications)

(Abelian connection)	$(R_g)^*\omega = \omega, \quad \Omega = \mathrm{d}\omega.$
(Local potential)	$A = s^*\omega, \quad A' = A + g^{-1}\mathrm{d}g.$
(Field strength)	$F = s^*\Omega = \mathrm{d}A, \quad F' = F, \quad \mathrm{d}F = 0.$
(Charge q line bundle)	$E_q = P \times_{\rho_q} \mathbb{C}, \quad \nabla^{(q)} = \mathrm{d} + (\rho_q)_*(A).$

3 Spinor bundles and the Dirac operator on Minkowski space

From now on we fix the 4-dimensional Minkowski space

$$M := \mathbb{R}^4, \qquad \eta = \mathrm{diag}(1, -1, -1, -1),$$

equipped with its standard orientation (given by $dx^0 \wedge dx^1 \wedge dx^2 \wedge dx^3$) and the standard time-orientation (the vector field ∂_0 is future-directed). This section constructs the (Dirac) spinor bundle, its canonical connection, a natural fiberwise pairing, and the associated Dirac operator in an intrinsic way.

3.1 The orthonormal frame bundle of $(\mathbb{R}^4, \eta)$

Let $\mathbb{R}^{1,3}$ denote $\mathbb{R}^4$ equipped with the bilinear form η of signature $(1, 3)$. An *oriented, time-oriented orthonormal frame* at $x \in M$ is a linear isometry

$$u : \mathbb{R}^{1,3} \longrightarrow T_x M$$

preserving orientation and time-orientation. The set of such frames forms a principal $\mathrm{SO}^+(1, 3)$-bundle over M, denoted

$$\pi_{\mathrm{SO}} : P_{\mathrm{SO}}(M) \longrightarrow M,$$

often called the (proper orthochronous) orthonormal frame bundle.
Since TM is trivial and η has constant coefficients in the standard coordinates, there is a canonical global orthonormal frame

$$e_0 = \partial_0, \qquad e_j = \partial_j \ (j = 1, 2, 3),$$

with dual coframe

$$\theta^0 = dx^0, \qquad \theta^j = dx^j.$$

In particular, the principal bundle $P_{\mathrm{SO}}(M)$ is trivial:

$$P_{\mathrm{SO}}(M) \cong M \times \mathrm{SO}^+(1, 3).$$

3.2 The spin group and the spin structure

Let $\mathrm{Spin}^+(1, 3)$ be the connected spin group in signature $(1, 3)$, i.e. the double cover

$$\lambda : \mathrm{Spin}^+(1, 3) \longrightarrow \mathrm{SO}^+(1, 3).$$

(One may identify $\mathrm{Spin}^+(1, 3) \simeq \mathrm{SL}(2, \mathbb{C})$, though we will not rely on this.)

Definition 3.1 (Spin structure)**.** A *(Lorentzian) spin structure* on (M, η) is a principal $\mathrm{Spin}^+(1, 3)$-bundle $\pi_{\mathrm{Spin}} : P_{\mathrm{Spin}}(M) \to M$ together with a smooth bundle map $\Lambda : P_{\mathrm{Spin}}(M) \to P_{\mathrm{SO}}(M)$ covering id_M, such that

$$\Lambda(p \cdot g) = \Lambda(p) \cdot \lambda(g), \qquad \forall\, p \in P_{\mathrm{Spin}}(M),\ g \in \mathrm{Spin}^+(1, 3),$$

and each fiber map $\Lambda_x : P_{\mathrm{Spin}}(M)_x \to P_{\mathrm{SO}}(M)_x$ is the standard 2-fold covering.

On $M = \mathbb{R}^4$, the spin structure is (canonically) trivial: choosing the global orthonormal frame above identifies

$$P_{\mathrm{Spin}}(M) \cong M \times \mathrm{Spin}^+(1, 3), \qquad \Lambda(x, g) = (x, \lambda(g)).$$

3.3 The Dirac spinor bundle and Clifford multiplication

Fix once and for all a *Dirac spin representation*

$$\kappa : \mathrm{Spin}^+(1,3) \longrightarrow GL(\Sigma), \qquad \Sigma \cong \mathbb{C}^4,$$

together with an extension of κ to a representation of the complexified Clifford algebra $\mathrm{Cl}^{\mathbb{C}}_{1,3} \simeq \mathrm{Mat}_4(\mathbb{C})$. (Equivalently: choose a faithful complex representation $\rho : \mathrm{Cl}^{\mathbb{C}}_{1,3} \to \mathrm{End}(\Sigma)$ such that κ is the restriction of ρ to $\mathrm{Spin}^+(1,3) \subset \mathrm{Cl}^{\times}_{1,3}$.)

Definition 3.2 (Spinor bundle)**.** The (Dirac) *spinor bundle* on M is the associated complex vector bundle

$$S \;:=\; P_{\mathrm{Spin}}(M) \times_{\kappa} \Sigma \;\longrightarrow\; M.$$

A section $\psi \in \Gamma(S)$ is a *Dirac spinor field*.

Because $P_{\mathrm{Spin}}(M)$ is trivial, so is S:

$$S \cong M \times \Sigma.$$

We stress, however, that the intrinsic definitions below do not depend on choosing this trivialization.

Clifford bundle and Clifford action. Let $\mathrm{Cl}(T^*\mathbb{R}^4, \eta)$ denote the real Clifford algebra bundle of (T^*M, η), characterized by being generated fiberwise by T^*_xM with relations

$$\alpha\beta + \beta\alpha = 2\,\eta^{-1}(\alpha,\beta)\,\mathrm{Id}, \qquad \alpha, \beta \in T^*_xM.$$

Write $\mathrm{Cl}(T^*\mathbb{R}^4, \eta) \otimes_{\mathbb{R}} \mathbb{C}$ for its complexification. A spin structure canonically provides a bundle map (the *Clifford action*)

$$c : \mathrm{Cl}(T^*\mathbb{R}^4, \eta) \otimes_{\mathbb{R}} \mathbb{C} \longrightarrow \mathrm{End}(S).$$

Restricting to $T^*M \subset \mathrm{Cl}(T^*\mathbb{R}^4, \eta)$ gives the Clifford multiplication

$$c : T^*M \otimes S \longrightarrow S, \qquad (\alpha, \psi) \longmapsto c(\alpha)\psi.$$

In any local oriented orthonormal coframe $\{\theta^a\}_{a=0}^3$ (so $\eta^{-1}(\theta^a, \theta^b) = \eta^{ab}$), set

$$\gamma^a := c(\theta^a) \in \mathrm{End}(S),$$

so that the Clifford relations become

$$\gamma^a\gamma^b + \gamma^b\gamma^a = 2\,\eta^{ab}\,\mathrm{Id}. \tag{6}$$

In the standard global coframe $\theta^\mu = dx^\mu$, the γ^μ are constant endomorphisms of $\Sigma \cong \mathbb{C}^4$ once a spin frame is chosen; this is the usual "gamma-matrix" picture.

3.4 The spin connection on S

Let ∇^{LC} be the Levi–Civita connection of η on TM (torsion-free and metric-compatible). As recalled in Section 1, ∇^{LC} is equivalent to a principal connection on the $\mathrm{SO}^+(1,3)$-bundle $P_{\mathrm{SO}}(M)$, whose connection form we denote by

$$\omega^{LC} \in \Omega^1(P_{\mathrm{SO}}(M); \mathfrak{so}(1,3))\,.$$

A spin structure lifts ω^{LC} uniquely to a principal connection on $P_{\mathrm{Spin}}(M)$:

$$\omega^{\mathrm{Spin}} \in \Omega^1(P_{\mathrm{Spin}}(M); \mathfrak{spin}(1,3))\,,$$

characterized by compatibility with the double cover Λ (via the induced Lie algebra map $\lambda_* : \mathfrak{spin}(1,3) \to \mathfrak{so}(1,3)$):

$$\lambda_* \circ \omega^{\mathrm{Spin}} = \Lambda^* \omega^{LC}.$$

Finally, since $S = P_{\mathrm{Spin}}(M) \times_\kappa \Sigma$ is an associated bundle, ω^{Spin} induces a covariant derivative (Section 1)

$$\nabla^S : \Gamma(S) \longrightarrow \Gamma(T^*M \otimes S),$$

called the *spin connection*.

Remark 3.3 (Characterization by Clifford compatibility)**.** The spin connection ∇^S is uniquely characterized by the following two properties:

1. it is compatible with Clifford multiplication:

$$\nabla^S_X\big(c(\alpha)\psi\big) = c(\nabla^{LC}_X \alpha)\psi + c(\alpha)\nabla^S_X \psi,$$

   for all vector fields X, 1-forms α, and spinor fields ψ;

2. it is induced by (and hence compatible with) the metric connection ∇^{LC} on TM.

Local formula. Let $\{e_a\}$ be a local oriented orthonormal frame with dual coframe $\{\theta^a\}$. Write the Levi–Civita connection 1-forms $\omega^a{}_b \in \Omega^1(M)$ by

$$\nabla^{LC} e_a = \omega^b{}_a \, e_b, \qquad \omega_{ab} + \omega_{ba} = 0 \quad (\omega_{ab} := \eta_{ac}\omega^c{}_b).$$

In a local spin frame, the induced spin connection takes the standard form

$$\nabla^S = \mathrm{d} + \frac{1}{4}\,\omega_{ab}\,\gamma^a\gamma^b. \tag{7}$$

(Here d is the trivial connection in the chosen local spin frame; the right-hand side is the usual expression in terms of the spin representation.)

Specialization to Minkowski space. In the standard global orthonormal frame $e_\mu = \partial_\mu$ on $M = \mathbb{R}^4$, one has $\omega_{ab} = 0$, hence

$$\nabla^S = \mathrm{d} \qquad \text{in the corresponding global spin trivialization } S \cong M \times \Sigma.$$

Equivalently, $\nabla^S_{\partial_\mu}\psi = \partial_\mu \psi$ for $\psi : M \to \Sigma$.

3.5 The Dirac pairing

For later variational formulas, we need a nondegenerate pairing on spinors that is invariant under $\mathrm{Spin}^+(1,3)$ and interacts well with Clifford multiplication.

On the Dirac representation space $\Sigma \cong \mathbb{C}^4$ there exists a (unique up to scale) $\mathrm{Spin}^+(1,3)$-invariant nondegenerate sesquilinear form h with the property that Clifford multiplication by real covectors is symmetric with respect to h. Concretely, after choosing a standard gamma-matrix realization satisfying

$$(\gamma^0)^\dagger = \gamma^0, \qquad (\gamma^j)^\dagger = -\gamma^j \quad (j = 1,2,3),$$

one may take

$$h(\psi, \phi) := \psi^\dagger \gamma^0 \, \phi, \tag{8}$$

which is invariant under the $\mathrm{Spin}^+(1,3)$-action and is nondegenerate (but indefinite).

This induces a fiberwise pairing on the bundle S (still denoted h),

$$h_x : S_x \times S_x \to \mathbb{C}, \qquad x \in M,$$

and allows one to define the *Dirac adjoint* $\bar\psi$ of a spinor field ψ as the section of the dual bundle S^* corresponding to $h(\psi, \cdot)$ (in a trivialization, $\bar\psi = \psi^\dagger\gamma^0$).

Remark 3.4 (Metric compatibility)**.** The spin connection ∇^S is compatible with h:

$$X\big(h(\psi,\phi)\big) = h(\nabla^S_X\psi,\phi) + h(\psi,\nabla^S_X\phi),$$

for all vector fields X and spinor fields ψ,ϕ.

3.6 The Dirac operator

Definition 3.5 (Geometric Dirac operator)**.** Let $\{e_a\}_{a=0}^3$ be any local oriented orthonormal frame and $\{\theta^a\}$ its dual coframe. The (Lorentzian) *Dirac operator* on (M,η) is the first-order differential operator

$$\not{D} : \Gamma(S) \longrightarrow \Gamma(S), \qquad \not{D}\psi := \sum_{a=0}^{3} c(\theta^a)\,\nabla^S_{e_a}\psi = \sum_{a=0}^{3} \gamma^a\,\nabla^S_{e_a}\psi.$$

The definition is independent of the choice of orthonormal frame, since a change of frame is a local $\mathrm{SO}^+(1,3)$-gauge transformation that lifts to $\mathrm{Spin}^+(1,3)$ and leaves the Clifford contraction invariant.

Remark 3.6. The Clifford module is used here in defining the Dirac operator, which is further used for defining the QED Lagrangian. The reason we can use the Clifford module is that the representation κ we used here extends to a Clifford module. If a representation that can not be extended to a Clifford module, extra care is needed in constructing the kinetic operator. In other words, the Clifford module structure helps us build the kinetic operator hence the Lagrangian. To define the matter field itself we only need to use the spin structure, which doesn't depend on the existence of the Clifford module.

Standard coordinates on $\mathbb{R}^4$. In the standard global orthonormal frame $e_\mu = \partial_\mu$ on Minkowski space, $\nabla^S = \mathrm{d}$ and thus

$$\not{D}\psi = \gamma^\mu\,\partial_\mu\psi, \tag{9}$$

where $\gamma^\mu = c(dx^\mu)$ satisfy the Clifford relations (6). In physics conventions one often uses $i\not{D}$ in the Dirac equation and in the Lagrangian; we will insert the factor of i at that stage.

Remark 3.7 (Flatness)**.** On $(\mathbb{R}^4,\eta)$ the Levi–Civita curvature vanishes, hence the spin connection is flat and $\not{D}$ has constant coefficients in a global inertial frame.

4 Spinors as relativistic matter fields and the spin-$\frac{1}{2}$ property

Section 3 constructed the (Dirac) spinor bundle $S \to (\mathbb{R}^4, \eta)$ as an associated bundle to the spin frame bundle. This section explains why *sections of S model relativistic spin-$\frac{1}{2}$ degrees of freedom* and how the half-integer spin is visible directly from the representation-theoretic input (the Dirac representation of $\mathrm{Spin}^+(1,3)$).

4.1 Relativistic fields as associated bundles

A classical relativistic "field with internal degrees of freedom" is, geometrically, a section of an associated vector bundle: one specifies

- a principal bundle encoding the relevant symmetry (here the local Lorentz symmetry, i.e. the spin frame bundle $P_{\mathrm{Spin}}(M)$),
- a representation of the structure group on a model fiber,
- and one forms the associated bundle.

In our case, the local Lorentz group is $\mathrm{SO}^+(1,3)$, whose double cover $\mathrm{Spin}^+(1,3)$ acts on the Dirac spin space $\Sigma \simeq \mathbb{C}^4$ via the Dirac spin representation κ. Thus the spinor bundle

$$S \;=\; P_{\mathrm{Spin}}(M) \times_\kappa \Sigma$$

is precisely the geometric object whose local descriptions are Σ-valued functions glued together by $\mathrm{Spin}^+(1,3)$-valued transition functions.

Remark 4.1 (Local change of spin frame $\Leftrightarrow$ gauge transformation)**.** Let $s : U \to P_{\mathrm{Spin}}(M)$ be a local spin frame. A spinor field $\psi \in \Gamma(S|_U)$ is represented by a function $\psi_s : U \to \Sigma$ via $\psi(x) = [s(x), \psi_s(x)]$. If $s' = s \cdot g$ with $g : U \to \mathrm{Spin}^+(1,3)$, then

$$\psi_{s'}(x) = \kappa\big(g(x)^{-1}\big)\, \psi_s(x).$$

This is the direct analogue of $U(1)$ gauge transformations for charged fields (Section 2), except that the structure group is now $\mathrm{Spin}^+(1,3)$.

4.2 Poincaré covariance and the spinorial transformation law

On Minkowski space the global spacetime symmetry is the (proper orthochronous) Poincaré group

$$\mathrm{ISO}^+(1,3) \;\cong\; \mathbb{R}^{1,3} \rtimes \mathrm{SO}^+(1,3),$$

and its double cover

$$\widetilde{\mathrm{ISO}}^+(1,3) \;\cong\; \mathbb{R}^{1,3} \rtimes \mathrm{Spin}^+(1,3)$$

acts naturally on spinors. In the standard global spin trivialization $S \cong \mathbb{R}^4 \times \Sigma$, a *global* Lorentz transformation $\Lambda \in \mathrm{SO}^+(1,3)$ with a chosen lift $g \in \mathrm{Spin}^+(1,3)$ (so $\lambda(g) = \Lambda$) acts on a spinor field by

$$(\Lambda \cdot \psi)(x) \;:=\; \kappa(g)\, \psi(\Lambda^{-1} x), \tag{10}$$

and a translation by $a \in \mathbb{R}^{1,3}$ acts by $(a \cdot \psi)(x) := \psi(x - a)$. Equation (10) is the standard "spinor transformation law": it combines the orbital action on the base point with the spin action on the fiber.

Remark 4.2 (Why the spin group appears)**.** If one only used $\mathrm{SO}^+(1,3)$ on the fiber, one would obtain tensor fields (vectors, forms, ...). Spinors are different: under a 2π spatial rotation, they pick up a minus sign. This forces the relevant group acting on the fiber to be the double cover $\mathrm{Spin}^+(1,3)$ rather than $\mathrm{SO}^+(1,3)$.

4.3 Infinitesimal Lorentz transformations and the generators

Let $\{\theta^a\}$ be an oriented orthonormal coframe and $\gamma^a = c(\theta^a)$ the Clifford action on S as in Section 3, so that

$$\gamma^a\gamma^b + \gamma^b\gamma^a = 2\,\eta^{ab}\,\mathrm{Id}.$$

The Dirac representation of the Lorentz Lie algebra $\mathfrak{so}(1,3)$ on Σ can be described intrinsically in terms of Clifford multiplication. Define the endomorphisms

$$\Sigma^{ab} \;:=\; \frac{1}{4}[\gamma^a,\gamma^b] \;\in\; \mathrm{End}(S), \qquad \Sigma^{ab} = -\Sigma^{ba}. \tag{11}$$

These satisfy the $\mathfrak{so}(1,3)$ commutation relations and provide the infinitesimal spin action. Concretely, for an antisymmetric parameter $\omega_{ab} = -\omega_{ba}$, the corresponding (local) spin action is given by the exponential

$$\psi \longmapsto \exp\Big(\frac{1}{2}\,\omega_{ab}\,\Sigma^{ab}\Big)\,\psi, \tag{12}$$

which is the bundle-level avatar of the Dirac representation.

Remark 4.3 (Relation to the spin connection formula)**.** In curved backgrounds one has locally $\nabla^S = \mathrm{d} + \frac{1}{4}\omega_{ab}\gamma^a\gamma^b$ from (7). The same matrices Σ^{ab} control how spinors respond to infinitesimal frame rotations/boosts: the spin connection is precisely the device that makes derivatives transform covariantly under local Lorentz changes of frame. On Minkowski space in an inertial frame, $\omega_{ab} = 0$ and $\nabla^S = \mathrm{d}$.

4.4 Spatial rotations and the appearance of spin $\frac{1}{2}$

Fix the standard future-directed unit timelike vector $e_0 = \partial_0$. The subgroup of $\mathrm{SO}^+(1,3)$ that fixes e_0 is (canonically) SO(3), acting as spatial rotations on $\mathrm{span}\{e_1, e_2, e_3\}$. Its preimage inside $\mathrm{Spin}^+(1,3)$ is $\mathrm{Spin}(3) \cong \mathrm{SU}(2)$:

$$\mathrm{Spin}(3) \;\cong\; \mathrm{SU}(2) \;\longrightarrow\; \mathrm{SO}(3) \quad \text{is the familiar double cover.}$$

Restricting the Dirac representation κ to this SU(2) subgroup yields the *spin content* of the field.

Decomposition under rotations. Under $\mathrm{SU}(2) \cong \mathrm{Spin}(3)$, the Dirac spin space decomposes as

$$\Sigma \;\cong\; \mathbb{C}^2 \oplus \mathbb{C}^2,$$

where each $\mathbb{C}^2$ is the *fundamental* (spin-$\frac{1}{2}$) representation of SU(2). Equivalently: a Dirac spinor contains two copies of a Pauli spinor as far as spatial rotations are concerned.

The 2π-rotation sign. The element $-1 \in \mathrm{SU}(2)$ maps to the identity rotation in SO(3), but in the fundamental representation it acts by $-\,\mathrm{Id}$ on $\mathbb{C}^2$. Consequently, in the Dirac representation one has

$$\kappa(-1) = -\,\mathrm{Id}_\Sigma.$$

Thus a full 2π spatial rotation sends a spinor field to minus itself:

$$\psi \longmapsto -\,\psi.$$

This is the hallmark of half-integer spin.

Spin operators and Pauli matrices (in a standard gamma realization). Choose a gamma-matrix realization in which

$$(\gamma^0)^\dagger = \gamma^0, \qquad (\gamma^j)^\dagger = -\gamma^j \quad (j = 1, 2, 3),$$

as in Section 3. Define the *(Hermitian) spin operators* by

$$S^k \;:=\; \frac{i}{2}\,\varepsilon^{kij}\,\Sigma^{ij} \qquad (k = 1, 2, 3), \tag{13}$$

where Σ^{ij} are as in (11) and ε^{kij} is the Levi–Civita symbol (with $\varepsilon^{123} = +1$). In the standard Dirac basis these are block-diagonal:

$$S^k \;=\; \frac{1}{2}\begin{pmatrix} \sigma^k & 0 \\ 0 & \sigma^k \end{pmatrix},$$

where σ^k are the Pauli matrices. In particular,

$$\vec{S}^{\,2} := (S^1)^2 + (S^2)^2 + (S^3)^2 = \frac{3}{4}\,\mathrm{Id}_\Sigma,$$

which is the algebraic statement that the intrinsic angular momentum is $s(s+1)$ with $s = \frac{1}{2}$.

Remark 4.4 (What this means physically)**.** In quantum theory, the rotation group acts on one-particle states by a unitary representation. The above shows that the internal degrees of freedom furnished by a Dirac spinor transform as spin-$\frac{1}{2}$: the two-dimensional fundamental representation of SU(2) under spatial rotations. This is the origin of the "spin up / spin down" states of the electron.

4.5 Chirality and the role of Dirac spinors

Although the electron is massive (and we will introduce its mass term later), it is conceptually helpful to note an additional structure that exists already at the level of representations. Define the chirality operator

$$\gamma^5 \;:=\; i\,\gamma^0\gamma^1\gamma^2\gamma^3. \tag{14}$$

It satisfies $(\gamma^5)^2 = \mathrm{Id}$ and anticommutes with each γ^a. Hence it yields a decomposition of the Dirac spin space into left- and right-handed (Weyl) subspaces:

$$\Sigma = \Sigma_L \oplus \Sigma_R, \qquad \Sigma_{L/R} := \ker(\gamma^5 \mp \mathrm{Id}), \qquad P_{L/R} := \frac{1}{2}(\mathrm{Id} \pm \gamma^5).$$

Under $\mathrm{Spin}^+(1,3) \cong \mathrm{SL}(2,\mathbb{C})$, Σ_L and Σ_R are the two inequivalent two-dimensional Weyl representations (often denoted $(\frac{1}{2}, 0)$ and $(0, \frac{1}{2})$), which are exchanged by parity. A *Dirac* spinor combines both chiralities and is therefore the natural choice when one wants a parity-symmetric description and, later, a Lorentz-invariant mass term.

Summary for later sections

At this stage we have only encoded the *spin* degrees of freedom:

- The electron field will be modeled (before turning on charge) as a section $\psi \in \Gamma(S)$ of the Dirac spinor bundle.
- The fact that $\kappa|_{\mathrm{Spin}(3)}$ contains the fundamental SU(2) representation, and that a 2π rotation acts by $-\mathrm{Id}$, is the geometric origin of spin-$\frac{1}{2}$.
- In Section 5 we will tensor S with a charge-q complex line bundle E_q and use the induced connection on $S \otimes E_q$ to implement minimal coupling to electromagnetism.

5 Minimal coupling via the product principal bundle and the Maxwell–Dirac system

In Sections 2–4 we described:

- the electromagnetic field as a principal U(1)-connection ω on a principal U(1)-bundle $P \to M$ (on $M = \mathbb{R}^4$ we take $P = M \times \mathrm{U}(1)$), and
- the spin-$\frac{1}{2}$ degrees of freedom via the spin principal bundle $P_{\mathrm{Spin}}(M) \to M$ and the associated Dirac spinor bundle $S = P_{\mathrm{Spin}}(M) \times_\kappa \Sigma$.

This section packages these two structures into a *single* principal bundle with structure group $\mathrm{Spin}^+(1,3) \times \mathrm{U}(1)$. In this language, "minimal coupling" is not a separate prescription: it is simply the functorial induced connection on an associated bundle.
Throughout, $M = \mathbb{R}^4$ with Minkowski metric $\eta = \mathrm{diag}(1, -1, -1, -1)$.

5.1 The product principal bundle $P_{\mathrm{Spin}}(M) \times_M P$

Let

$$\pi_{\mathrm{Spin}} : P_{\mathrm{Spin}}(M) \to M, \qquad \pi_{\mathrm{U}(1)} : P \to M$$

be the spin principal bundle and the electromagnetic U(1) principal bundle. Their fiber product over M is the smooth manifold

$$\mathcal{P} \;:=\; P_{\mathrm{Spin}}(M) \times_M P \;=\; \{(p_{\mathrm{Spin}}, p) \in P_{\mathrm{Spin}}(M) \times P :\; \pi_{\mathrm{Spin}}(p_{\mathrm{Spin}}) = \pi_{\mathrm{U}(1)}(p)\}.$$

It is a principal bundle over M with structure group

$$G_{\mathrm{tot}} := \mathrm{Spin}^+(1,3) \times \mathrm{U}(1),$$

with right action (consistent with Section 1)

$$(p_{\mathrm{Spin}}, p) \cdot (g, z) \;:=\; (p_{\mathrm{Spin}} \cdot g,\; p \cdot z). \tag{15}$$

A local section of $\mathcal{P}$ on $U \subset M$ is equivalently a pair of local sections

$$\hat{s} = (s_{\mathrm{Spin}}, s_{\mathrm{U}(1)}), \qquad s_{\mathrm{Spin}} : U \to P_{\mathrm{Spin}}(M), \quad s_{\mathrm{U}(1)} : U \to P.$$

A change of local section is given by a map

$$\hat{g} : U \to \mathrm{Spin}^+(1,3) \times \mathrm{U}(1), \qquad \hat{g} = (g, h),$$

so that $\hat{s}' = \hat{s} \cdot \hat{g}$ means $s'_{\mathrm{Spin}} = s_{\mathrm{Spin}} \cdot g$ and $s'_{\mathrm{U}(1)} = s_{\mathrm{U}(1)} \cdot h$.

Remark 5.1 (Two types of "gauge transformations")**.** The $\mathrm{Spin}^+(1,3)$ part corresponds to changes of local spin frame (local Lorentz gauge), while the U(1) part is the electromagnetic gauge freedom of Section 2. The coupled field equations will be U(1) gauge invariant, and the spin-frame covariance is built in by the use of the spin principal bundle.

5.2 The product connection: a canonical way to "add" connections

Let

$$\omega^{\mathrm{Spin}} \in \Omega^1\big(P_{\mathrm{Spin}}(M); \mathfrak{spin}(1,3)\big)$$

be the spin principal connection induced from the Levi–Civita connection (Section 3), and let

$$\omega \in \Omega^1(P; \mathfrak{u}(1))$$

be the electromagnetic principal connection (Section 2).
Denote by

$$\mathrm{pr}_{\mathrm{Spin}} : \mathcal{P} \to P_{\mathrm{Spin}}(M), \qquad \mathrm{pr}_{\mathrm{U}(1)} : \mathcal{P} \to P$$

the projections. Define the $\big(\mathfrak{spin}(1,3) \oplus \mathfrak{u}(1)\big)$-valued 1-form

$$\hat{\omega} \;:=\; \mathrm{pr}_{\mathrm{Spin}}^{*}\,\omega^{\mathrm{Spin}} \;\oplus\; \mathrm{pr}_{\mathrm{U}(1)}^{*}\,\omega \;\in\; \Omega^{1}\big(\mathcal{P};\; \mathfrak{spin}(1,3) \oplus \mathfrak{u}(1)\big). \tag{16}$$

Proposition 5.2 (Product principal connection)**.** *The form $\hat{\omega}$ is a principal connection on the principal* $(\mathrm{Spin}^{+}(1,3) \times \mathrm{U}(1))$*-bundle $\mathcal{P} \to M$. Equivalently, its horizontal distribution is*

$$H_{\hat{\omega}} = \ker(\mathrm{pr}_{\mathrm{Spin}}^{*}\,\omega^{\mathrm{Spin}}) \;\cap\; \ker(\mathrm{pr}_{\mathrm{U}(1)}^{*}\,\omega).$$

Curvature. Let Ω^{Spin} and Ω be the curvatures of ω^{Spin} and ω. Since the Lie algebra of a product group is the direct sum and the bracket is componentwise (and $\mathfrak{u}(1)$ is abelian), the curvature of $\hat{\omega}$ splits with no cross terms:

$$\hat{\Omega} \;:=\; \mathrm{d}\hat{\omega} + \tfrac{1}{2}[\hat{\omega} \owedge \hat{\omega}] \;=\; \mathrm{pr}_{\mathrm{Spin}}^{*}\,\Omega^{\mathrm{Spin}} \;\oplus\; \mathrm{pr}_{\mathrm{U}(1)}^{*}\,\Omega. \tag{17}$$

Thus the electromagnetic field strength is precisely the $\mathfrak{u}(1)$-component of the total curvature.

Remark 5.3 (Minkowski specialization)**.** On $(\mathbb{R}^4, \eta)$, $\Omega^{\mathrm{Spin}} = 0$ and in a global inertial spin frame one has $\omega^{\mathrm{Spin}} = 0$. The only nontrivial curvature component in (17) is the electromagnetic one.

5.3 Charged Dirac spinors as an associated bundle of $\mathcal{P}$

Fix an integer $q \in \mathbb{Z}$ and the character (Section 2)

$$\rho_q : \mathrm{U}(1) \to \mathrm{U}(1) \subset \mathbb{C}^{\times}, \qquad \rho_q(z) = z^{q}.$$

Let $\kappa : \mathrm{Spin}^{+}(1,3) \to GL(\Sigma)$ be the Dirac representation (Section 4). Define the tensor product representation

$$\kappa_q \;:=\; \kappa \otimes \rho_q : \mathrm{Spin}^{+}(1,3) \times \mathrm{U}(1) \to GL(\Sigma \otimes \mathbb{C}), \qquad \kappa_q(g,z)(\psi \otimes \alpha) = \kappa(g)\psi \otimes z^{q}\alpha.$$

(We keep $\Sigma \otimes \mathbb{C}$ to make the two factors visible; one may canonically identify it with Σ since $\mathbb{C}$ is one-dimensional.)

Definition 5.4 (Bundle of charged Dirac spinors)**.** The *charged Dirac bundle* is the associated bundle

$$W_q \;:=\; \mathcal{P} \times_{\kappa_q} (\Sigma \otimes \mathbb{C}) \;\longrightarrow\; M.$$

A classical charged electron field is a section $\Psi \in \Gamma(W_q)$.

Proposition 5.5 (Identification with $S \otimes E_q$)**.** *Let $S = P_{\mathrm{Spin}}(M) \times_{\kappa} \Sigma$ and $E_q = P \times_{\rho_q} \mathbb{C}$. There is a canonical vector bundle isomorphism*

$$W_q \;\cong\; S \otimes E_q,$$

given fiberwise by

$$\big[(p_{\mathrm{Spin}}, p),\; \psi \otimes \alpha\big] \longmapsto [p_{\mathrm{Spin}}, \psi] \;\otimes\; [p, \alpha].$$

Remark 5.6 (Local transformation law)**.** If $\hat{s} = (s_{\mathrm{Spin}}, s_{\mathrm{U}(1)})$ is a local section of $\mathcal{P}$ on U and $\Psi \in \Gamma(W_q|_U)$ is represented by $\psi : U \to \Sigma \otimes \mathbb{C}$ via $\Psi(x) = [\hat{s}(x), \psi(x)]$, then under $\hat{s}' = \hat{s} \cdot (g,h)$ one has

$$\psi'(x) = \kappa_q\big(g(x)^{-1}, h(x)^{-1}\big)\,\psi(x) = \kappa(g(x)^{-1})\,\psi(x) \;\otimes\; h(x)^{-q}.$$

This simultaneously encodes the spin-frame transformation and the $U(1)$ phase transformation.

5.4 Minimal coupling = the induced connection from the product connection

By Section 1, the principal connection $\hat{\omega}$ on $\mathcal{P}$ induces a covariant derivative on *every* associated bundle, in particular on W_q.

Definition 5.7 (Minimally coupled covariant derivative)**.** Let $\hat{\omega}$ be as in (16). The *minimally coupled* covariant derivative on W_q is the induced connection

$$\nabla^{W_q} : \Gamma(W_q) \to \Gamma(T^*M \otimes W_q)$$

associated to the representation κ_q.

Local formula. Let $\hat{s} = (s_{\mathrm{Spin}}, s_{\mathrm{U}(1)})$ be a local section on $U \subset M$ and set

$$A^{\mathrm{Spin}} := s_{\mathrm{Spin}}^* \omega^{\mathrm{Spin}} \in \Omega^1\big(U; \mathfrak{spin}(1,3)\big), \qquad A := s_{\mathrm{U}(1)}^* \omega \in \Omega^1(U; \mathfrak{u}(1)).$$

Then the pulled-back total gauge potential is

$$\hat{A} := \hat{s}^* \hat{\omega} = (A^{\mathrm{Spin}}, A) \in \Omega^1\big(U; \mathfrak{spin}(1,3) \oplus \mathfrak{u}(1)\big).$$

If $\Psi = [\hat{s}, \psi]$ with $\psi : U \to \Sigma \otimes \mathbb{C}$, then (Section 1)

$$\nabla^{W_q} \psi \;=\; \mathrm{d}\psi \;+\; (\kappa_q)_*(\hat{A})\, \psi. \tag{18}$$

Since $(\kappa \otimes \rho_q)_*$ splits on the direct sum Lie algebra, one has

$$(\kappa_q)_*(X, Y) \;=\; \kappa_*(X) \otimes \mathrm{Id} \;+\; \mathrm{Id} \otimes (\rho_q)_*(Y), \qquad X \in \mathfrak{spin}(1,3),\; Y \in \mathfrak{u}(1). \tag{19}$$

Thus (18) becomes

$$\nabla^{W_q} \psi = \mathrm{d}\psi + \big(\kappa_*(A^{\mathrm{Spin}}) \otimes \mathrm{Id}\big)\psi + \big(\mathrm{Id} \otimes (\rho_q)_*(A)\big)\psi.$$

Proposition 5.8 (Equivalence with the tensor-product connection)**.** *Under the identification $W_q \cong S \otimes E_q$ of Proposition 5.5, the induced connection ∇^{W_q} agrees with the tensor-product connection*

$$\nabla^S \otimes \mathrm{Id}_{E_q} \;+\; \mathrm{Id}_S \otimes \nabla^{(q)}.$$

Minkowski inertial frame. On $M = \mathbb{R}^4$ in a global inertial spin frame one has $A^{\mathrm{Spin}} = 0$ and $\nabla^S = \mathrm{d}$. Writing $A = i\mathcal{A}$ with $\mathcal{A} \in \Omega^1(M; \mathbb{R})$ (Section 2), and using $(\rho_q)_*(A) = qA = iq\,\mathcal{A}$, one gets

$$\nabla^{W_q} = \mathrm{d} + iq\,\mathcal{A}. \tag{20}$$

5.5 The coupled Dirac operator

Clifford multiplication on S extends to $W_q \cong S \otimes E_q$ by acting trivially on E_q:

$$c_{W_q}(\alpha) := c(\alpha) \otimes \mathrm{Id}_{E_q}, \qquad \alpha \in T^*M.$$

In an orthonormal frame $\{e_a\}$ with dual $\{\theta^a\}$, write $\gamma^a = c(\theta^a)$ as before.

Definition 5.9 (Minimally coupled Dirac operator)**.** The *minimally coupled Dirac operator* on W_q is

$$\not{D}_{\mathcal{A}} \Psi \;:=\; \sum_{a=0}^{3} c_{W_q}(\theta^a)\, \nabla^{W_q}_{e_a} \Psi = \sum_{a=0}^{3} \gamma^a\, \nabla^{W_q}_{e_a} \Psi.$$

In standard Minkowski coordinates and an inertial spin frame, (20) gives

$$\not{D}_{\mathcal{A}} \psi = \gamma^\mu \big(\partial_\mu + iq\,\mathcal{A}_\mu\big)\psi, \qquad \psi : \mathbb{R}^4 \to \Sigma \cong \mathbb{C}^4. \tag{21}$$

5.6 Intrinsic QED Lagrangian and the Euler–Lagrange equations

Pairings. Let h_S be the Dirac pairing on S (Section 3) and let $\langle\cdot,\cdot\rangle$ be the standard Hermitian metric on $\mathbb{C}$, which is U(1)-invariant. Then $h_S \otimes \langle\cdot,\cdot\rangle$ is $\mathrm{Spin}^+(1,3)\times \mathrm{U}(1)$-invariant on $\Sigma\otimes\mathbb{C}$, hence it descends to a fiberwise pairing on W_q, denoted h_{W_q}. We write $\bar\Psi$ for the Dirac adjoint (the W_q^*-valued field determined by h_{W_q}), which in an inertial trivialization is $\bar\psi = \psi^\dagger\gamma^0$.

Electromagnetic curvature. Using the real potential $\mathcal{A}$ (so $A = i\mathcal{A}$), define the real field strength

$$\mathcal{F} := \mathrm{d}\mathcal{A} \in \Omega^2(M;\mathbb{R}).$$

Equivalently, the $\mathfrak{u}(1)$-valued curvature is $F = \mathrm{d}A = i\mathcal{F}$, and it is the $\mathfrak{u}(1)$-component of $\hat\Omega$ in (17).

Definition 5.10 (QED Lagrangian density (Dirac + Maxwell))**.** Fix a mass parameter $m \geq 0$ (natural units $\hbar = c = 1$). The QED Lagrangian 4-form is

$$\mathcal{L}_{\mathrm{QED}}(\Psi,\mathcal{A}) := \mathcal{L}_{\mathrm{Dirac}}(\Psi,\mathcal{A}) + \mathcal{L}_{\mathrm{Max}}(\mathcal{A}),$$

where

$$\mathcal{L}_{\mathrm{Dirac}}(\Psi,\mathcal{A}) := \left(\frac{i}{2}\sum_{a=0}^{3}\Big(h_{W_q}\big(\Psi,\ \gamma^a\nabla^{W_q}_{e_a}\Psi\big) - h_{W_q}\big(\nabla^{W_q}_{e_a}\Psi,\ \gamma^a\Psi\big)\Big) - m\,h_{W_q}(\Psi,\Psi)\right)\mathrm{vol}_\eta, \quad (22)$$

$$\mathcal{L}_{\mathrm{Max}}(\mathcal{A}) := -\frac{1}{2}\,\mathcal{F}\wedge\star\mathcal{F}. \quad (23)$$

Remark 5.11 (Why the product viewpoint is conceptually clean)**.** Everything in (22) is built from: (i) the total principal connection $\hat\omega$ on the total symmetry bundle $\mathcal{P}$, (ii) the associated representation κ_q, (iii) the $\mathrm{Spin}^+(1,3)\times\mathrm{U}(1)$-invariant pairing on the model fiber $\Sigma\otimes\mathbb{C}$, and (iv) the Lorentzian metric η (via vol_η and $\star$). No extra coupling term is introduced by hand.

Action and variations. On $M = \mathbb{R}^4$ we take variations with compact support (or impose boundary conditions on a bounded domain) and define

$$\mathcal{S}[\Psi,\mathcal{A}] := \int_M \mathcal{L}_{\mathrm{QED}}(\Psi,\mathcal{A}).$$

Dirac equation (variation in the matter field)

Varying with respect to $\bar\Psi$ yields the Euler–Lagrange equation

$$i\,\not{D}_{\mathcal{A}}\Psi - m\,\Psi = 0. \quad (24)$$

In standard coordinates and an inertial spin frame this reads

$$\Big(i\,\gamma^\mu(\partial_\mu + iq\,\mathcal{A}_\mu) - m\Big)\psi = 0, \qquad \psi:\mathbb{R}^4\to\mathbb{C}^4. \quad (25)$$

Maxwell equations (variation in the U(1) **connection)**

Varying with respect to the potential $\mathcal{A}$ gives the inhomogeneous Maxwell equation. Define the current 1-form $j^\flat\in\Omega^1(M)$ by

$$j^\flat(X) \ :=\ q\ \mathrm{Re}\Big(h_{W_q}\big(\Psi,\ c_{W_q}(X^\flat)\Psi\big)\Big), \qquad X\in\Gamma(TM), \quad (26)$$

where $X^\flat = \eta(X, \cdot)$. Then the Maxwell system is

$$\mathrm{d}\mathcal{F} = 0, \qquad \mathrm{d} \star \mathcal{F} = \star j^\flat. \tag{27}$$

In standard coordinates this becomes

$$\partial_{[\lambda}\mathcal{F}_{\mu\nu]} = 0, \qquad \partial_\mu \mathcal{F}^{\mu\nu} = j^\nu, \qquad j^\nu = q\,\bar{\psi}\gamma^\nu\psi, \quad \bar{\psi} = \psi^\dagger\gamma^0. \tag{28}$$

Remark 5.12 (Current conservation)**.** Taking d of $\mathrm{d} \star \mathcal{F} = \star j^\flat$ gives $\mathrm{d} \star j^\flat = 0$, i.e. $\partial_\mu j^\mu = 0$. Equivalently, this follows from U(1) gauge invariance of the action (Noether).

Cheat sheet (flat Minkowski, inertial spin frame)

Total principal bundle:	$\mathcal{P} = P_{\mathrm{Spin}}(M) \times_M P, \quad G_{\mathrm{tot}} = \mathrm{Spin}^+(1,3) \times \mathrm{U}(1).$
Total connection:	$\hat{\omega} = \mathrm{pr}^*_{\mathrm{Spin}}\,\omega^{\mathrm{Spin}} \oplus \mathrm{pr}^*_{\mathrm{U}(1)}\,\omega.$
Charged spinor bundle:	$W_q = \mathcal{P} \times_{\kappa\otimes\rho_q} (\Sigma \otimes \mathbb{C}) \cong S \otimes E_q.$
Minimal coupling:	$D_\mu = \partial_\mu + iq\,\mathcal{A}_\mu, \quad \mathcal{F}_{\mu\nu} = \partial_\mu\mathcal{A}_\nu - \partial_\nu\mathcal{A}_\mu.$
Field equations:	$(i\gamma^\mu D_\mu - m)\psi = 0, \quad \partial_\mu\mathcal{F}^{\mu\nu} = q\,\bar{\psi}\gamma^\nu\psi, \quad \partial_{[\lambda}\mathcal{F}_{\mu\nu]} = 0.$

6 General matter representations, intrinsic parity, and spin selection by equations of motion

This section records a representation-theoretic "user manual" for the product principal bundle viewpoint of Section 5, and explains how parity enters as additional structure not visible inside $\mathrm{Spin}^+(1,3)$. We also discuss the Proca system as an example where the equation of motion imposes constraints that remove "irrelevant" spin components.
Throughout we remain on Minkowski space $M = \mathbb{R}^4$ with metric $\eta = \mathrm{diag}(1,-1,-1,-1)$.

6.1 Generic matter fields from the total principal bundle

Recall from Section 5 the *total* principal bundle

$$\mathcal{P} \;:=\; P_{\mathrm{Spin}}(M) \times_M P \;\longrightarrow\; M, \qquad G_{\mathrm{tot}} := \mathrm{Spin}^+(1,3) \times \mathrm{U}(1),$$

where $P_{\mathrm{Spin}}(M)$ is the spin principal bundle and P is the electromagnetic U(1)-bundle. Minimal coupling was formulated as: choose a representation of G_{tot}, form the associated bundle, and use the induced connection from the product principal connection.

Irreducible representations of $\mathrm{Spin}^+(1,3)$

One convenient way to package the finite-dimensional complex representation theory is via the standard labeling by pairs

$$(j_L, j_R) \in \tfrac{1}{2}\mathbb{N} \times \tfrac{1}{2}\mathbb{N}.$$

Concretely, using the Lie algebra isomorphism

$$\mathfrak{so}(1,3) \otimes_{\mathbb{R}} \mathbb{C} \;\cong\; \mathfrak{sl}(2,\mathbb{C}) \oplus \mathfrak{sl}(2,\mathbb{C}),$$

an irreducible representation is determined by specifying an $\mathfrak{sl}(2,\mathbb{C})$-spin j_L on the first factor and an $\mathfrak{sl}(2,\mathbb{C})$-spin j_R on the second factor.
On the bundle side, it is often convenient to use the chiral splitting of the Dirac bundle (Section 4): the chirality operator

$$\gamma^5 := i\,\gamma^0\gamma^1\gamma^2\gamma^3$$

defines projectors $P_{L/R} = \frac{1}{2}(\mathrm{Id} \pm \gamma^5)$ and hence subbundles

$$S_L := P_L S, \qquad S_R := P_R S, \qquad S \cong S_L \oplus S_R.$$

Then the irreducible $\mathrm{Spin}^+(1,3)$-bundles corresponding to (j_L, j_R) can be realized (up to canonical identifications) as the "spinor-tensor" bundles

$$\mathcal{V}_{(j_L,j_R)} \;:=\; \mathrm{Sym}^{2j_L}(S_L) \;\otimes\; \mathrm{Sym}^{2j_R}(S_R). \tag{29}$$

(For example, $(\frac{1}{2},0)$ is S_L, $(0,\frac{1}{2})$ is S_R, and the Dirac bundle corresponds to $(\frac{1}{2},0)\oplus(0,\frac{1}{2})$.)

Adding electric charge

Fix $q \in \mathbb{Z}$ and the U(1) character $\rho_q(z) = z^q$ (Section 2), with associated line bundle

$$E_q := P \times_{\rho_q} \mathbb{C}.$$

Given *any* (possibly reducible) complex representation

$$\rho : \mathrm{Spin}^+(1,3) \to GL(V),$$

define the corresponding "Lorentz bundle"

$$\mathcal{V} := P_{\mathrm{Spin}}(M) \times_\rho V.$$

A *charged* matter bundle of charge q is then

$$\mathcal{V}_q \;:=\; \mathcal{V} \otimes E_q. \tag{30}$$

Equivalently (and canonically in the product-bundle viewpoint), $\mathcal{V}_q$ is the associated bundle

$$\mathcal{V}_q \;\cong\; \mathcal{P} \times_{\rho\otimes\rho_q} (V \otimes \mathbb{C}), \tag{31}$$

where $(g, z) \in \mathrm{Spin}^+(1,3) \times \mathrm{U}(1)$ acts by $\rho(g) \otimes \rho_q(z)$. A classical (charged) matter field is a section

$$\Phi \in \Gamma(\mathcal{V}_q).$$

Minimal coupling (connection)

Let $\hat{\omega}$ be the product principal connection on $\mathcal{P}$ from Section 5. By Section 1, $\hat{\omega}$ induces a covariant derivative on every associated bundle, in particular on $\mathcal{V}_q$:

$$\nabla^{\mathcal{V}_q} : \Gamma(\mathcal{V}_q) \to \Gamma(T^*M \otimes \mathcal{V}_q).$$

This is the intrinsic formulation of "minimal coupling": nothing is added beyond the product principal connection and the choice of representation.

6.2 Parity as extra structure: intrinsic parity operators

Parity on Minkowski space

Let

$$p : M \to M, \qquad p(x^0, x^1, x^2, x^3) = (x^0, -x^1, -x^2, -x^3)$$

be the parity diffeomorphism. Its differential is the Lorentz transformation $\Lambda_P = \mathrm{diag}(1, -1, -1, -1) \in O(1,3)$, which has determinant -1 and therefore does *not* lie in $\mathrm{SO}^+(1,3)$. Consequently, parity is not encoded inside $\mathrm{Spin}^+(1,3)$ and must be specified separately.

The parity automorphism on $\mathrm{Spin}^+(1,3)$

Conjugation by Λ_P defines an automorphism of the proper Lorentz Lie algebra $\mathfrak{so}(1,3)$, and hence induces an (outer) automorphism of $\mathrm{Spin}^+(1,3)$, which we denote by

$$\alpha_P : \mathrm{Spin}^+(1,3) \to \mathrm{Spin}^+(1,3).$$

On the representation labels, α_P exchanges the two chiral spins:

$$(j_L, j_R) \;\longleftrightarrow\; (j_R, j_L).$$

Equivalently, for any representation $\rho_{(j_L,j_R)}$ one has

$$\rho_{(j_L,j_R)} \circ \alpha_P \;\simeq\; \rho_{(j_R,j_L)}.$$

Definition 6.1 (Intrinsic parity operator on a representation)**.** Let $\rho : \mathrm{Spin}^+(1,3) \to GL(V)$ be a complex representation. An *intrinsic parity operator* for ρ is a linear isomorphism $\Pi : V \to V$ such that

$$\Pi\, \rho(g)\, \Pi^{-1} = \rho(\alpha_P(g)) \qquad \forall\, g \in \mathrm{Spin}^+(1,3), \tag{32}$$

together with the normalization

$$\Pi^2 = \mathrm{Id}_V. \tag{33}$$

Remark 6.2 (Existence and "intrinsic parity sign")**.** If ρ is irreducible and $\rho \simeq \rho \circ \alpha_P$ (equivalently $j_L = j_R$ for an irrep), then any intertwiner Π is unique up to a scalar by Schur's lemma, and the condition $\Pi^2 = \mathrm{Id}$ reduces this freedom to $\Pi \mapsto \pm\Pi$. The choice of this sign is the *intrinsic parity*. If ρ is not parity-stable (e.g. (j_L, j_R) with $j_L \neq j_R$), then one typically forms the parity doublet $\rho \oplus (\rho \circ \alpha_P)$ and lets Π exchange the two summands.

Parity action on associated bundles and parity-even/odd fields

Let $\mathcal{V}_q$ be the charged matter bundle (30) built from ρ and q. Assume we have chosen an intrinsic parity operator Π for ρ. (Parity acts trivially on U(1) in standard QED, so ρ_q is not altered by parity.)
On Minkowski space we can use the standard global trivializations to write parity on sections as

$$(\mathsf{P}\Phi)(x) \;:=\; \Pi\,\Phi(p^{-1}x), \qquad \Phi \in \Gamma(\mathcal{V}_q). \tag{34}$$

A field configuration is called

$$\textit{parity-even} \text{ if } \; \mathsf{P}\Phi = \Phi, \qquad \textit{parity-odd} \text{ if } \; \mathsf{P}\Phi = -\Phi.$$

Remark 6.3 (Parity-even/odd is a constraint on configurations)**.** The condition $\mathsf{P}\Phi = \pm\Phi$ identifies values of Φ at x with values at the reflected point $p^{-1}x$. Thus it is a nonlocal restriction on the space of field configurations. A parity-symmetric *theory* does not require each configuration to be parity-even or parity-odd; rather, it requires that parity maps solutions to solutions (see the next subsection).

6.3 Example: electron field and the parity doublet $(\frac{1}{2}, 0) \oplus (0, \frac{1}{2})$

The Dirac representation is the parity doublet

$$(\tfrac{1}{2}, 0) \oplus (0, \tfrac{1}{2}),$$

corresponding to the Dirac bundle $S \cong S_L \oplus S_R$. The charge-q electron bundle is $W_q = S \otimes E_q$ (Section 5).
In a standard gamma-matrix realization, parity on Dirac spinors is implemented by

$$\Pi_D \;:=\; \eta_P\,\gamma^0, \qquad \eta_P \in \{\pm 1\}, \tag{35}$$

where the sign η_P is the intrinsic parity choice. One checks that γ^0 intertwines the Dirac representation with its parity twist, i.e. it satisfies (32) for $\rho = \kappa$ (the Dirac representation), and $\Pi_D^2 = \mathrm{Id}$ holds in standard conventions.

Parity swaps chirality. Because γ^0 anticommutes with γ^5, one has

$$\gamma^0 P_L = P_R \gamma^0, \qquad \gamma^0 P_R = P_L \gamma^0.$$

Thus parity exchanges the left- and right-handed components of a Dirac spinor.

Parity-even/odd constraint. Writing $\psi = \psi_L + \psi_R$ with $\psi_{L/R} = P_{L/R}\psi$, the parity action (34) with (35) reads

$$(\mathsf{P}\psi)(x) = \eta_P\,\gamma^0\,\psi(p^{-1}x). \tag{36}$$

The parity-even condition $\mathsf{P}\psi = \psi$ is therefore equivalent to the coupled constraints

$$\psi_L(x) = \eta_P\,\gamma^0\,\psi_R(p^{-1}x), \qquad \psi_R(x) = \eta_P\,\gamma^0\,\psi_L(p^{-1}x),$$

and similarly for the parity-odd condition $\mathsf{P}\psi = -\psi$ (with an overall sign change). This exhibits explicitly how parity eigenconditions relate the two chiral components.

6.4 Parity (non)invariance of the Lagrangian and parity breaking in the equations of motion

Parity invariance as a symmetry of the action

Let $\mathcal{L}(\Phi, \mathcal{A})$ be a Lagrangian density (a 4-form) built from a matter field $\Phi \in \Gamma(\mathcal{V}_q)$ and an electromagnetic potential $\mathcal{A} \in \Omega^1(M; \mathbb{R})$ (Section 2). Parity acts on $\mathcal{A}$ by pullback:

$$\mathcal{A} \longmapsto p^*\mathcal{A} \qquad (\text{equivalently } \mathcal{A}_0(t, \mathbf{x}) \mapsto \mathcal{A}_0(t, -\mathbf{x}),\ \mathcal{A}_i(t, \mathbf{x}) \mapsto -\mathcal{A}_i(t, -\mathbf{x})).$$

We say the action is *parity invariant* if

$$\int_M \mathcal{L}(\Phi, \mathcal{A}) \ = \ \int_M \mathcal{L}(\mathsf{P}\Phi,\ p^*\mathcal{A}) \tag{37}$$

for all compactly supported variations. In this case, the Euler–Lagrange equations are *parity covariant*: if $(\Phi, \mathcal{A})$ solves the equations of motion, then so does $(\mathsf{P}\Phi, p^*\mathcal{A})$.

QED is parity-even (vector-like coupling)

In Section 5 we used the same U(1) charge representation on both chiral pieces of the Dirac bundle, i.e. the U(1) factor acts by a scalar ρ_q times the identity on Σ. In this *vector-like* situation, the minimally coupled Dirac–Maxwell Lagrangian is parity invariant with the standard parity operator $\Pi_D = \eta_P \gamma^0$.

How parity can be broken by the Lagrangian

Even if the *field content* admits a parity operator, the *Lagrangian* may fail to respect it. Two common (and geometrically transparent) mechanisms are:

1. **Chiral gauge couplings.** Let $S \cong S_L \oplus S_R$ and suppose the U(1)-representation on the fiber is *not* a scalar: assign different charges to left and right,

   $$\rho_{\mathrm{U}(1)}(z)\big|_{S_L} = z^{q_L}, \qquad \rho_{\mathrm{U}(1)}(z)\big|_{S_R} = z^{q_R}.$$

   Then the minimally coupled covariant derivative takes the form

   $$D_\mu = \partial_\mu + i\,\mathcal{A}_\mu\,(q_L P_L + q_R P_R).$$

   Parity exchanges P_L and P_R, so parity invariance forces $q_L = q_R$. If $q_L \neq q_R$, the action (and hence the equations of motion) breaks parity.

2. **Parity-odd fermion bilinears.** Even with $q_L = q_R$, adding terms involving γ^5 breaks parity, e.g.

   $$\mathcal{L}_{\text{odd}} \ = \ m_5\,\bar\psi\, i\gamma^5\psi\ \mathrm{vol}_\eta \qquad \text{or} \qquad \mathcal{L}_{\text{axial}} \ = \ g_A\,\bar\psi\,\gamma^\mu\gamma^5\psi\ \mathcal{A}_\mu\ \mathrm{vol}_\eta,$$

   since γ^5 changes sign under parity (it is pseudoscalar).

Remark 6.4 (What "parity breaking in the EOM" means)**.** If (37) fails, parity will not map solutions to solutions in general. Equivalently, imposing a parity-even constraint $\mathsf{P}\Phi = \Phi$ on initial data will not be preserved by time evolution unless the equations are parity covariant.

6.5 The Proca system: removing irrelevant spins by constraints

A useful example of "spin selection by equations of motion" is a massive spin-1 field.

Bundle and representation content

A Lorentz vector (or covector) transforms in the $(\frac{1}{2},\frac{1}{2})$ representation. Geometrically, the corresponding bundle is TM or T^*M (built from $P_{\mathrm{Spin}}(M)$ via the vector representation). A *charged* massive spin-1 field of charge q is naturally modeled as a section

$$B \;\in\; \Omega^1(M;E_q) \;\cong\; \Gamma(T^*M\otimes E_q).$$

Under spatial rotations $\mathrm{Spin}(3)\cong SU(2)$, one has the decomposition

$$(\tfrac{1}{2},\tfrac{1}{2})\big|_{SU(2)} \;\cong\; 0\oplus 1,$$

so a priori a covector field contains both spin-0 and spin-1 components. The Proca equation imposes a constraint that removes the unwanted spin-0 component.

Gauge-covariant exterior derivative

Let $\nabla^{(q)}$ be the connection on E_q induced from the U(1) connection (Section 2). It extends to an exterior covariant derivative on E_q-valued forms:

$$D := d_{\nabla^{(q)}} : \Omega^k(M;E_q)\to\Omega^{k+1}(M;E_q).$$

Define the Proca field strength

$$H := DB \in \Omega^2(M;E_q).$$

Using the metric η and the Hermitian metric on E_q, define the formal adjoint

$$D^\dagger := -\star D\star : \Omega^{k+1}(M;E_q)\to\Omega^k(M;E_q).$$

(Equivalently, $D^\dagger$ is the coderivative associated to the minimally coupled connection.)

Proca equation and the constraint

The (minimally coupled) Proca equation for a mass $m>0$ is

$$D^\dagger H + m^2 B = 0, \qquad \text{where } H = DB. \tag{38}$$

In standard Minkowski coordinates this reads

$$D_\mu H^{\mu\nu} + m^2 B^\nu = 0, \qquad H_{\mu\nu} = D_\mu B_\nu - D_\nu B_\mu, \qquad D_\mu = \partial_\mu + iq\,\mathcal{A}_\mu.$$

In the free case (or, more generally, in situations where the commutator terms do not obstruct the argument), applying $D^\dagger$ to (38) yields the on-shell constraint

$$D^\dagger B = 0. \tag{39}$$

In flat coordinates and without coupling this is simply $\partial_\mu B^\mu = 0$. This constraint removes one component from the four-component covector field, leaving 3 propagating degrees of freedom, as expected for a massive spin-1 particle.

Spin selection. Representation-theoretically, (39) eliminates the spin-0 part in the $0\oplus 1$ decomposition under spatial rotations, leaving precisely the spin-1 sector. Thus the Proca equation enforces that the field propagates the correct spin.

Remark 6.5 (Massless limit and the photon)**.** Formally setting $m=0$ in (38) restores a gauge redundancy (in the uncoupled case). The reason is that when $m=0$, the Proca equation becomes $D^\dagger DB=0$. Since $D^2=0$, we have a redundancy of $B\to B+D\chi$. Hence the number of physical degrees of freedom drops from 3 to 2 because of this redundancy, matching the two photon polarizations. In the bundle language of Section 2, the massless spin-1 field is most naturally encoded by the U(1)-connection itself (or equivalently by its curvature F).